

# Hamdan Khan

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## EDUCATION

### Ontario Tech University

*Bachelor of Engineering (Honours) in Software Engineering*

Oshawa, Ontario

*Expected April 2027*

## TECHNICAL SKILLS

**Languages:** Java, Python, JavaScript, C, C++, SQL, TypeScript, PHP

**Frameworks:** React, Flask, Tailwind, Spring Boot, JUnit, CI/CD Pipelines

**Developer Tools:** Git, Bitbucket, Jira, Confluence, Docker, Figma, MySQL, GDB, Jenkins, Kubernetes

**Software Development:** Agile, SDLC, API Design, Version Control, Unit Testing, Data Structures, Algorithms, SOLID

## EXPERIENCE

### Protection, Control & Automation Engineering Co-op Student

Sep 2025 – Aug 2026

*Hydro One Networks Inc.*

*Ontario, Canada*

- Designed and executed countless end-to-end integration test cases validating real-time data flow between distributed embedded endpoints and centralized backend reporting systems
- Developed and deployed firmware logic on embedded devices to aggregate station-level signals and report alarm/fault states across **15+** distributed system components
- Built integration and regression test pipelines for real-time telemetry systems, achieving **85%** test coverage and reducing defect detection time by **35%**
- Diagnosed and resolved **20+** cross-layer communication failures by tracing data flow across firmware logic, network transmission, and backend processing services
- Validated bidirectional communication interfaces between control systems and field devices, ensuring reliable command dispatch and telemetry feedback across **30+** monitoring endpoints
- Supported high-availability operational dashboards by validating telemetry integrity and ensuring sub-second propagation of critical system events to control room operators

### Software Engineer Intern

Mar 2025 – Apr 2025

*Wouessi Digital*

*Toronto, Ontario*

- Built a TypeScript/Node.js backend to extract and process **7K+** federal IT contracts for benchmarking analysis
- Developed REST APIs to expose procurement data for downstream pipelines and dashboard integration
- Engineered a Jenkins CI pipeline to automate installs, linting, unit tests, Docker builds, and Kubernetes deployments
- Optimized parsing and filtering logic to support accurate and scalable insight generation from contract metadata

## PROJECTS

### Automated Rail System Log Analyzer | *Python, Shell Scripting, Git, OOP, NLP, Transformers*

- Designed an OOP-based Python parser to analyze and summarize **50+** simulated embedded system logs, achieving **100%** detection accuracy for warnings and errors
- Automated the analysis workflow using shell scripting, cutting log triage time by **70%** and simulating real-world DevOps efficiency
- Integrated BART transformer model for natural language summarization of logs, improving report clarity and operator readability by **80%**

### Hospital Laboratory System | *PHP, REST API, MySQL, JavaScript*

- Automated test ordering, result processing, and billing for hospital labs
- Designed a relational database schema with normalized MySQL tables for optimized queries
- Developed RESTful API endpoints to handle real-time updates and scheduling

### AI-Controlled 2D Racing Game | *Python, Pygame, Stable Baseline3*

- Implemented a Deep-Q learning network for AI-driven obstacle detection and pathfinding, improving decision-making by **50%**

## CERTIFICATIONS

**Software Development (BC):** SDLC, Cloud Computing, Security, API Development, Testing

**Harvard CS50X:** C Programming, OOP, Data Structures, Machine Learning, SQL, Full-Stack Web Development

**Harvard AI with Python:** Graph Search, Supervised and Reinforcement Learning

**Udemy Python BootCamp:** Web Scraping, GUI Development, OOP